

Project Blue Book

Special Report No. 14

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SUMMARY

Reports of unidentified aerial objects (popularly termed “flying saucers” or “flying discs”) have been received by the U.S. Air Force since mid-1947 from many and diverse sources. Although there was no evidence that the unexplained reports of unidentified objects constituted a threat to the security of the U.S., the Air Force determined that all reports of unidentified aerial objects should be investigated and evaluated to determine if “flying saucers” represented technological developments not known to this country.

In order to discover any pertinent trend or pattern inherent in the data, and to evaluate or explain any trend or pattern found, appropriate methods of reducing these data from reports of unidentified aerial objects to a form amenable to scientific appraisal were employed. In general, the original data upon which this study was based consisted of impressions and interpretations of apparently unexplainable events, and seldom contained reliable measurements of physical attributes. This subjectivity of the data presented a major limitation to the drawing of significant conclusions, but did not invalidate the application of scientific methods of study.

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The reports received by the U.S. Air Force on unidentified aerial objects were reduced to IBM punched-card abstracts of the data by means of logically developed forms and standardized evaluation procedures. Evaluation of sighting reports, a crucial step in the preparation of the data for statistical treatment, consisted of an appraisal of the reports and the subsequent categorization of the object or objects described in each report. A detailed description of this phase of the study stresses the careful attempt to maintain complete objectivity and consistency.

Analysis of the refined and evaluated data derived from the original reports of sightings consisted of (1) a systematic attempt to ferret out any distinguishing characteristics inherent in the data of any of their segments, (2) a concentrated study of any trend or pattern found, and (3) an attempt to determine the probability that any of the UNKNOWNNS represent observations of technological developments not known to this country.

The first step in the analysis of the data revealed the existence of certain apparent similarities between cases of objects definitely identified and those not identified. Statistical methods of testing when applied indicated a low probability that these apparent similarities were significant. An attempt to determine the probability that any of the UNKNOWNNS represented observations of technological developments not known to this country necessitated a thorough re-examination and re-evaluation of the cases of objects not originally identified; this led to the conclusion that this probability was very small.

The special study which resulted in this report (Analysis of Reports of Unidentified Aerial Objects, 5 May 1955) started in 1953. To provide the study group with a complete set of files, the information cut-off date was established as of the end of 1952. It will accordingly be noted that the statistics contained in all charts and tables in this report are terminated with the year 1952. In these charts, 3201 cases have been used.

As the study progressed, a constant program was maintained for the purpose of making comparisons between the current cases received after 1 January 1953, and those being used for the report. This was done in order that any change or significant trend which might arise from current developments could be incorporated in the summary of this report

PROJECT BLUE BOOK

The 1953 and 1954 cases show a general and expected trend of increasing percentages in the finally identified categories. They also show decreasing percentages in categories where there was insufficient information and those where the phenomena could not be explained. This trend had been anticipated in the light of improved reporting and investigating procedures.

Official reports on hand at the end of 1954 totaled 4834. Of these, 425 were produced in 1953 and 429 in 1954. These 1953 and 1954 individual reports (a total of 854), were evaluated on the same basis as were those received before the end of 1952. The results are as follows:

- *Balloons – 16%*
- *Aircraft – 20%*
- *Astronomical – 25%*
- *Other – 13%*
- *Insufficient Info – 17%*
- *Unknown – 9%*

As the study of the current cases progressed, it became increasingly obvious that if reporting and investigating procedures could be further improved, the percentages of those cases which contained insufficient information and those remaining unexplained would be greatly reduced. The key to a higher percentage of solutions appeared to be in rapid “on the spot” investigations by trained personnel. On the basis of this, a revised program was established by AF Reg. 200-2 Subject: “Unidentified Flying Objects Reporting” (Short Title: UFOB) dated 12 August 1954.

This new program, which had begun to show marked results before January 1955, provided primarily that the 4602d Air Intelligence Service Squadron (Air Defense Command) would carry out all field investigations. This squadron has sufficient units and is so deployed as to be able to arrive “on the spot” within a very short time after a report is received. After treatment by the 4602d AISS, all information is supplied to the Air Technical Intelligence Center for final evaluation. This cooperative program has resulted, since 1 January 1955, in reducing the insufficient information cases to 7% and the unknown cases to 3% of the totals.

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The period 1 January 1955 to 5 May 1955 accounted for 131 unidentified aerial object reports received. Evaluation percentages of these are as follows:

- Balloons – 26%*
- Aircraft – 21%*
- Astronomical – 23%*
- Other – 20%*
- Insufficient Info – 7%*
- Unknown – 3%*

All available data were included in this study which was prepared by a panel of scientists both in and out of the Air Force. On the basis of this study it is believed that all the unidentified aerial objects could have been explained if more complete observational data had been available. Insofar as the reported aerial objects which still remain unexplained are concerned, there exists little information other than the impressions and interpretations of their observers. As these impressions and interpretations have been replaced by the use of improved methods of investigation and reporting, and by scientific analysis, the number of unexplained cases has decreased rapidly towards the vanishing point.

Therefore, on the basis of this evaluation of the information, it is considered to be highly improbable that reports of unidentified aerial objects examined in this study represent observations of technological developments outside of the range of present-day scientific knowledge. It is emphasized that there has been a complete lack of any valid evidence of physical matter in any case of a reported unidentified aerial object.

INTRODUCTION

In June, 1947, Kenneth Arnold, a Boise, Idaho, businessman and private pilot, publicly reported the now-famous sighting of a chainlike formation of disc-shaped objects near Mount Rainier, Washington. Resulting newspaper publicity of this incident caught the public interest, and, shortly thereafter, a rash of reports of unidentified aerial objects spawned the term “flying saucers”. During the years since 1947, many reports of unidentified aerial objects have been received by the Air Force from many and diverse sources.

The unfortunate term “flying saucer”, or “flying disc”, because of its widespread and indiscriminate use, requires definition. Many definitions have been offered, one of the best being that originated by Dr. J. Allen Hynek, Director of the Emerson McMillin Observatory of The Ohio State University, who has taken a scientific interest in the problem of unidentified aerial objects since 1949. Dr. Hynek’s definition of the term is “any aerial phenomenon or sighting that remains unexplained to the viewer at least long enough for him to write a report about it” (1). Dr. Hynek, elaborating on his definition, says, “Each flying saucer, so defined,

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has associated with it a probable lifetime. It wanders in the field of public inspection like an electron in a field of ions, until 'captured' by an explanation which puts an end to its existence as a 'flying saucer'" (1).

This definition would be applicable to any and all of the sightings which remained unidentified throughout this study. However, the term "flying saucers" shall be used hereafter in this report to mean a novel, airborne phenomenon, a manifestation that is not a part of or readily explainable by the fund of scientific knowledge known to be possessed by the Free World. This would include such items as natural phenomena that are not yet completely understood, psychological phenomena, or intruder aircraft of a type that may be possessed by some source in large enough numbers so that more than one independent mission may have been flown and reported. Thus, these phenomena are of the type which should have been observed and reported more than once.

Since 1947, public interest in the subject of unidentified aerial objects fluctuated more or less within reasonable limits until the summer of 1952, when the frequency of reports of sightings reached a peak, possibly stimulated by several articles on the subject in leading popular magazines.

Early in 1952, the Air Force's cumulative study and analysis of reported sightings indicated that the majority of reports could be accounted for as misinterpretations of known objects (such as meteors, balloons, or aircraft), a few as the result of mild hysteria, and a very few as the result of unfamiliar meteorological phenomena and optical aberrations. However, a significant number of fairly complete reports by reliable observers remained unexplained. Although no evidence existed that unexplained reports of sightings constituted a physical threat to the security of the U.S., in March, 1952, the Air Force decided that all reports of unidentified aerial objects should be investigated and evaluated to determine if "flying saucers" represented technological developments not known to this country.

Originally, the problem involved the preparation and analysis of about 1,300 reports accumulated by the Air Force between 1947 and the end of March, 1952. During the course of the work, the number of reports submitted for analysis and evaluation more than tripled, the result of the unprecedented increase in observations during 1952. Accordingly, this study is based on a number of reports

PROJECT BLUE BOOK

considered to be large enough for a preliminary statistical analysis, approximately 4,000 reports.

This study was undertaken primarily to categorize the available reports of sightings and to determine the probability that any of the reports of unidentified aerial objects represented observations of “flying saucers”. With full cognizance of the quality of the data available for study, yet with an awareness of the proportions this subject has assumed at times in the public mind, this work was undertaken with all the seriousness accorded to a straightforward scientific investigation. In order to establish the probability that any of the reports of unidentified aerial objects represented observations of “flying saucers”, it was necessary to make an attempt to answer the question “What is a ‘flying saucer’?”. However, it must be emphasized that this was only incidental to the primary purpose of the study, the determination of the probability that any of the reports of unidentified aerial objects represented observations of “flying saucers”, as defined on Page 1.

The basic technique for this study consisted of reducing the available data to a form suitable for mechanical manipulation, a prerequisite for the application of preliminary statistical methods. One of International Business Machine Corporation’s systems was chosen as the best available mechanical equipment.

The reduction of data contained in sighting reports into a form suitable for transfer to IBM punched cards was extremely difficult and time consuming.

For this study a panel of consultants was formed, consisting of both experts within and outside ATIC. During the course of the work, guidance and advice were received from the panel. The professional experience available from the panel covered major scientific fields and numerous specialized fields.

All records and working papers of this study have been carefully preserved in an orderly fashion suitable for ready reference. These records include condensations of all individual sighting reports, and the IBM cards used in various phases of the study.

ORIGIN AND NATURE OF DATA

Reports of sightings were received by the U.S. Air Force from a representative cross section of the population of the U.S., and varied widely in completeness and quality. Included were reports from reputable scientists, housewives, farmers, students, and technically trained members of the Armed Forces. Reports varied in length from a few sentences stating that a "flying saucer" had been sighted, to those containing thousands of words, including description, speculation, and advice on how to handle the "problem of the 'flying saucers'". Some reports were of high quality, conservative, and as complete as the observer could make them; a few originated from people confined to mental institutions. A critical examination of the reports revealed, however, that a high percentage of them was submitted by serious people, mystified by what they had seen and motivated by patriotic responsibility.

Three principal sources of reports were noted in the preliminary review of the data. The bulk of the data arrived at ATIC through regular military channels, from June, 1947, until the middle of 1952.

A second type of data consisted of letters reporting sightings sent by civilian observers directly to ATIC. Most of these direct communications were dated subsequent to April 30, 1952, and are believed to be the result of a suggestion by a popular magazine that future reports be directed to the Air Technical Intelligence Center. As could be expected, a large number of letters was received following this publicity.

A third type of data was that contained in questionnaire forms completed by the observer himself. A questionnaire form, developed during the course of this study, was mailed by ATIC to a selected group of writers of direct letters with the request that the form be completed and returned. Approximately 1,000 responses were received by ATIC.

In general, the data were subjective, consisting of qualified estimates of physical characteristics rather than of precise measurements. Furthermore, most of the reports were not reduced to written form immediately. The time between sighting and report varied from one day to several years. Both of these factors introduced an element of doubt concerning the validity of the original data, and increased its subjectivity. This was intensified by the recognized inability of the

average individual to estimate speeds, distances, and sizes of objects in the air with any degree of accuracy. In spite of these limitations, methods of statistical analysis of such reports in sufficiently large groups are valid. The danger lies in the possibility of forgetting the subjectivity of the data at the time that conclusions are drawn from the analysis. It must be emphasized, again and again, that any conclusions contained in this report are based NOT on facts, but on what many observers thought and estimated the true facts to be.

Altogether, the data for this study consisted of approximately 4,000 reports of sightings of unidentified aerial objects. The majority were received through military channels or in the form of observer-completed questionnaires; a few were accepted in the form of direct letters from unquestionably reliable sources. Sightings made between June, 1947, and December, 1952, were considered for this study. Sightings alleged to have occurred prior to 1947 were not considered, since they were not reported to official sources until after public interest in “flying saucers” had been stimulated by the popular press.

REDUCTION OF DATA TO MECHANIZED COMPUTATION FORM

As received by the Air Technical Intelligence Center, the sighting reports were not in a form suitable for even a quasi-scientific study. A preliminary review of the data indicated the need for standardized interrogation procedures and supplemental forms for the reduction of currently held and subsequently acquired data to a form amenable to scientific appraisal.

The plan for reduction of the data to usable form consisted of a program of development comprising four major steps:

- (1) a systematic listing of the factors necessary to evaluate the observer and his report, and to identify the unknown object observed;
- (2) a standard scheme for the transfer of data to a mechanized computation system;
- (3) an orderly means of relating the original data to all subsequent forms; and
- (4) a consistent procedure for the identification of the phenomenon described by the original data.

QUESTIONNAIRE

The first reports received by ATIC varied widely in completeness and quality. Air Force Letter 200-5(2) and Air Force Form 112(1) were attempts to fix responsibility for and improve the quality of the reports of sightings. To coordinate past efforts and to provide standardization for the future, it was imperative to develop a questionnaire form listing the factors necessary for evaluation of the observer and his report, and identification of the unknown objects. In addition, it was decided that such a questionnaire should be designed to serve as an interrogator's guide, and as a form for the observer himself to complete when personal interrogation was not possible or practicable.

Ideally, a questionnaire for the purposes required should contain questions pertaining to all technical details considered to be essential for the statistical approach, and should serve to obtain a maximum of information from the average individual who had made a sighting in the past or would be likely to be reporting sightings in the future. Besides these discrete facts, an integrated written description of a sighting would be required, thus enabling the reported facts of the sighting to be corroborated. Also, a narrative description might allow subtle questions to be answered concerning the observer's ability, such as indirect questions that would reveal his reasoning ability, suggestibility, and general mental attitude. As a whole, then, the information contained in a questionnaire should make possible the classification and evaluation of the sighting, the rating of the observer, the probability of accuracy of reported facts, and the identification of what was reported by the observer as unidentified.

During the course of this project, three questionnaire forms were developed, each intended to be an improved revision of the one preceding. The improvements were suggested and confirmed by members of the panel of consultants connected with this project.

The original form was evolved by the panel of consultants as their first work on this project. It was intended to allow the start of the reduction of reports to discrete data, and was immediately subjected to extensive review and revision by the panel. The revised (second) form was subjected to a trial test before adoption. ATIC sent a

copy to observers reporting sightings, with the request that the form be completed and returned. Of the first 300 questionnaires returned during July and August, 1952, 168 were analyzed by a consulting psychologist. On the basis of this analysis, plus the experience gained in working with past reports, the final form of the questionnaire — the U.S. Air Force Technical Information Sheet — was evolved. Copies of the three forms of the questionnaire, in the order of their development, are shown as Exhibits B1, B2, and B3 in Appendix B.

In order to implement the transcription of data from past sighting reports, each succeeding form was put to use as soon as it was developed and approved. Accordingly, experience was obtained with each form in relation to past data, an important factor in the improvement of the quality and completeness of the later reports included in this study.

CODING SYSTEM AND WORK SHEET

The reduction of non-numerical data to numerical form is mandatory in the machine handling of data. Thus, the selection of the IBM punched-card system for analysis of data forced the adoption of a master coding plan. Since it was impracticable to transfer detailed data of an exact nature from the questionnaire to the IBM card, an intermediate transfer form, coordinated with the master code, was necessary.

The master coding plan was evolved during the early stages of the preliminary analysis of data, and was reviewed by the panel of consultants before use. It was recognized that this system of coding would be the heart of the analysis, that is, the completeness of the facility for translation of data could make or break the study. Accordingly, every conceivable factor that might influence the identification of unidentified aerial objects was included, together with a wide range of variations within each factor. The original coding system (with minor corrections) was used throughout the translation of the original data with marked success. A copy of this system, called CODES, is enclosed as Exhibit B4, Appendix B.

To facilitate the preparation of the punched-card abstract, an intermediate form called the WORK SHEET (later, the CARD BIBLE) was developed. Referenced to both the data from the questionnaire and the system of report identification, the WORK SHEET permitted an orderly transcription of data simultaneously by

several people. In conjunction with the CODES, the WORK SHEET was used during the reduction of the original data to code form necessary for transfer to punched cards. A sample is included as Exhibit B5, Appendix B.

After the analysis was under way, it became apparent that the mechanics of machine processing could be improved by incorporating in the IBM card system group classifications of certain factors requiring more than one column for discrete expression. In addition, the inclusion of certain data relating to the evaluation and bearing of the sun with respect to the observer was considered necessary. Finally, a critical examination of certain segments of the data indicated the need for the definition of a new factor relating to the maneuvers of the object or objects sighted. Prior to the start of the analytical study, it had been assumed that a combination of stated factors would, by inference, define the maneuver pattern.

All these additions have been incorporated in a revised set of CODES and CARD BIBLE that are illustrated as Exhibits B6 and B7, Appendix B. However, at the time that the maneuver factor was determined to be critical, it was physically impracticable to make the required definitions and re-evaluate the original data. Therefore, no code for maneuverability has been included in the CODES, CARD BIBLE, or IBM cards.

IDENTIFICATION OF WORKING PAPERS

The actual reduction of data to IBM punched-card form presented a problem of mass transfer of figures by several workers. Recognizing that an orderly system of relating the original data to the questionnaire, the WORK SHEET, and the IBM card was imperative, a scheme of SERIAL NUMBERS was developed to answer this need.

The first data consisted of a series of letter-file folders identified by the year and location of the sighting or sightings they contained. The number of reports of sightings in a single folder varied from 1 to over 20. Under these conditions, there was a great possibility for incorrect transcription of data, duplication of transcription, or misplacement of intermediate forms. Further, it was considered desirable to relate all sightings of the same object or objects to one another. The concept of a four-digit serial number (major), followed by a two-digit subserial

number (minor), was adequate to fulfill these requirements.

To expedite handling of the data, temporary serial numbers were assigned until each report had been evaluated and the phenomenon had been placed in a category of identification. The use of temporary serial numbers permitted the consolidation of duplicate reports from apparently diverse sources, such as a teletype message and an Air Force Form 112. However, this consolidation was made ONLY when it could be proved conclusively that the sources of the two documents were one and the same. Factors of the observer's location, date and time of observation, description of the phenomenon, and finally, the name of the observer were considered. In this manner, the assignment of major serial and minor subserial numbers in continuous series was made only to the reports accepted for the statistical study. It is believed that the reports accepted represent unique and unduplicated instances of sightings.

In the establishment of the serial-number system, it was necessary to define certain terms, so that a standard interpretation could be achieved. The terms and corresponding definitions were:

OBSERVER – Any witness reporting to a proper authority that he had seen unidentified aerial objects.

SIGHTING – The report or group of reports of the same observed phenomenon that remained unidentified to the observer or observers, at least until reported.

SINGLE OBSERVATION – A SIGHTING consisting of a single report from (1) one OBSERVER with no knowledge of additional OBSERVERS of the same phenomenon, or (2) a group of witnesses of the same phenomenon, each cognizant of the others. The witness who made the report is called a SINGLE OBSERVER.

MULTIPLE OBSERVATION – A SIGHTING consisting of several reports from OBSERVERS of the same phenomenon who were cognizant of each other. The witnesses who made reports are called MULTIPLE OBSERVERS.

ALL SIGHTINGS – (1) The group of reports consisting of one report for each OBSERVER, including both SINGLE and MULTIPLE OBSERVERS. (2) The questionnaire, work sheet, and IBM card representing the report from each OBSERVER – in other words, the representation of each report accepted for the statistical study.

UNIT SIGHTINGS – (1) The group of reports consisting of one report for each SIGHTING, including all the reports of SINGLE OBSERVATIONS and the one most representative report from each MULTIPLE OBSERVATION. (2) The questionnaire, work sheet, and IBM card representing the report for each

SIGHTING accepted for the statistical study.

A major serial number (four digits) was assigned to each sighting, segregating the year of occurrence by selection of limits for each year, as follows:

0001 to 0500 reserved for 1947

0501 to 1000 reserved for 1948

1001 to 1500 reserved for 1949

1501 to 2000 reserved for 1950

2001 to 2500 reserved for 1951

2501 to 4900 reserved for 1952

While this scheme would serve to identify any individual sighting, identification of each report and its subsequent forms was necessary. The minor subserial numbers (two digits) fulfilled this requirement. For all SINGLE OBSERVATIONS, a major serial number followed by two (2) zeros, for example, 2759.00, was sufficient identification. For MULTIPLE OBSERVATIONS, the major serial number followed by a series of two-digit numbers ranging from 00 to 99 was used to identify the individual reports. In general, the most complete report from the most reliable observer of that MULTIPLE OBSERVATION was identified with the .00 subserial number.

As an example, a MULTIPLE OBSERVATION consisting of six sighting reports would have the following serial numbers:

1132.00 representing the best report and observer

1132.01 representing an additional observer

1132.02 representing an additional observer

1132.03 representing an additional observer

1132.04 representing an additional observer

1132.05 representing an additional observer

During the course of the transcription of the data to machine card form, it became obvious that certain reports could have been independent observations of the same phenomenon. So, if the presentation of an analysis based on one report for each sighting was valid (the concept of UNIT SIGHTINGS), a presentation of an analysis based on one report for each phenomenon should be valid also. Further, the examination of data relating to the actual number of

phenomena was considered to be the proper basis for assessing the probability of technological developments outside the range of present-day scientific knowledge. Therefore, a designation of OBJECT SIGHTINGS was established, with the following definition:

OBJECT SIGHTING – (1) The group of reports consisting of one report for each phenomenon. (2) The questionnaire, work sheet, and IBM card representing a report for each phenomenon accepted for the statistical study.

In brief review, ALL SIGHTINGS refer to all reports, UNIT SIGHTINGS refer to actual sightings, and OBJECT SIGHTINGS refer to the assumed number of phenomena.

It must be recognized that the process of identifying OBJECT SIGHTINGS was deductive, while that for UNIT SIGHTINGS was definitive. A conservative approach was adopted in the determination of OBJECT SIGHTINGS, using the factors of date and time of observations, location of observers, duration of observations, and range, bearing, track direction, and identification of the phenomena. Any error of selection of OBJECT SIGHTINGS will tend to be in the direction of reducing the actual number of phenomena observed (several instances of UNIT SIGHTINGS that might be one OBJECT SIGHTING were noted, but the evidence was not conclusive enough to justify consolidation of the reports).

Following the determination of OBJECT SIGHTINGS, a series of serial numbers, called the INCIDENT SERIAL NUMBERS, was established to facilitate any future study of a specific object sighting. Each reported sighting that relates to an OBJECT SIGHTING received the same incident serial number, a four-digit code paralleling the major serial number series.

For machine manipulation, it was desirable to be able to select the sample of cards (all reports, all sightings, or all phenomena) to be included in a particular study. The concept of a SIGHTING IDENTIFICATION NUMBER was evolved to fill this desire. Using one column of the IBM card, and the correlated working papers, the code for this function was developed. Multiple punching eliminated the need to use several columns for discrete expression of the variations. Selection of the proper number in this column thus permitted selection of the desired sample of cards.

EVALUATION OF INDIVIDUAL REPORTS

Evaluation of sighting reports was recognized as a crucial step in the preparation of data for statistical treatment; inconsistent evaluations would have invalidated any conclusions to be derived from this study. A method of evaluation was, therefore, determined simultaneously with the development of the questionnaire, the coding system, and the work sheet. It is emphasized that all phases of evaluation, even including the tedious preparation of the original data for statistical treatment, were entrusted only to selected, specially qualified scientists and engineers.

Evaluation consisted of a standardized procedure to be followed for: (1) the deduction of discrete facts from data which depended on human impressions rather than scientific measurements, (2) the rating of the observer and his report as determined from available information, and (3) the determination of the probable identification of the phenomenon observed. Categories of identification, established upon the basis of previous experience, were as follows:

- Balloon
- Astronomical
- Aircraft
- Light phenomenon
- Birds
- Clouds, dust, etc.
- Insufficient information
- Psychological manifestations
- Unknown
- Other

The first step in evaluation, the deduction of discrete facts from subjective data, required certain calculations based on the information available in the sighting report. An example was the finding of the approximate angular velocity and acceleration of the object or objects sighted. Care was taken during this phase of the work to insure against the deduction of discrete facts not warranted by the original data. Thus, even though there was a complete lack of any valid evidence consisting of physical matter in any case of a reported unidentified aerial object, this was not assumed to be *prima facie* evidence that “flying saucers” did not exist.

In those cases in which an attempt to reduce the information to a factual level failed completely, the report was eliminated from further consideration, and thus not included in the statistical analysis. About 800 reports of sightings were eliminated or rejected in this manner. Most of these reports were rejected because they were extremely nebulous; the rest were rejected because they contained highly conflicting statements.

The second step in evaluation, the rating of the observer and his report, logically followed the first step, the reduction of the data to usable form. Ratings were assigned on the basis of the following factors of information, considered in relation to one another:

1. The experience of the observer, deduced from his occupation, age, and training;
2. The consistency among the separate portions of the description of the sighting;
3. The general quality and completeness of the report;
4. Consideration of the observer's fact-reporting ability and attitude, as disclosed by his manner of describing the sighting.

In cases in which insufficient information was available to make a judgment of the observer or report, none was made, but the report was accepted for the statistical study.

The third step in the process of evaluation, the attempted identification of the object or objects sighted, was done twice, first by the individual who made the transcription of the data (the preliminary identification), and later (the final identification) by a conference of four persons, two representatives from ATIC and two from the panel of consultants. Although representatives of ATIC participated in making the final identifications, it must be emphasized that any previous identification of a sighting made by ATIC was not introduced or referred to in any way.

In the coding system, the choices provided for final identifications were based on ATIC's previous experience in analysis of the data. They had found that the majority of sightings could be classified as misinterpretations of common objects or natural phenomena. Accordingly, categories for objects most frequently present in the air were provided. Balloons, aircraft, astronomical bodies (such as meteors), birds, and clouds or dust were recognized as major categories. The less frequent, but

common objects, such as kites, fireworks, flares, rockets, contrails, and meteorological phenomena like small tornadoes, were collected into a category called OTHER. A separate category for the uncommon natural phenomena associated with light reflections or refractions, such as mirages, sun dogs, inversion-layer images, and distortions caused by airborne ice, was established with the title of LIGHT PHENOMENON. Categories for INSUFFICIENT INFORMATION, PSYCHOLOGICAL MANIFESTATIONS, and UNKNOWN were provided for the sightings that could not be fitted into the preceding identifications. An explanation of their use follows:

INSUFFICIENT INFORMATION – This identification category was assigned to a report when, upon final consideration, there was some essential item of information missing, or there was enough doubt about what data were available to disallow identification as a common object or some natural phenomenon. It is emphasized that this category of identification was not used as a convenient way to dispose of what might be called “poor unknowns,” but as a category for reports that, perhaps, could have been one of several known objects or natural phenomena. No reports identified as INSUFFICIENT INFORMATION contain authenticated facts or impressions concerning the sighting that would prevent its being identified as a known object or phenomenon.

PSYCHOLOGICAL MANIFESTATIONS – This identification category was assigned to a report when, although it was well established that the observer had seen something, it was also obvious that the description of the sighting had been overdrawn. Religious fanaticism, a desire for publicity, or an over-active imagination were the most common mental aberrations causing this type of report.

UNKNOWN – This designation in the identification code was assigned to those reports of sightings wherein the description of the object and its maneuvers could not be fitted to the pattern of any known object or phenomenon.

For the purposes of this study, two groups of identifications were recognized, the KNOWNs (including all identification categories except the UNKNOWNs) and the UNKNOWNs.

All possible identifications provided in the code system, except INSUFFICIENT INFORMATION and UNKNOWN, could be assigned according to two degrees of certainty, designated “Certain” and “Doubtful”.

A “Certain” identification indicated a minimum amount of doubt regarding the validity of the evaluation. By “rule-of-thumb” reasoning, the probability of the identification being correct was better than 95 per cent. A “Doubtful” identification indicated that the choice was less positive, but that there was a better than even chance of being correct.

It is emphasized again that, as was true for other phases of evaluation, preliminary and final identification was entrusted only to scientists and engineers who, in addition to their broad scientific background, had received instruction, where necessary, in specialized subjects. The panel of consultants provided background information for this instruction. Many of the cases representing unusual features or maneuvers were submitted to and discussed with various members of the panel of consultants prior to the final identification.

Consistency in the application of the knowledge necessary for making identifications was maintained by frequent collaboration among the personnel involved, and systematic spot checks of the work. In addition to the general fund of knowledge required to identify satisfactorily a reported unidentified aerial object, an attempt was made to correlate specific data such as flight plans of aircraft, records of balloon releases, weather conditions, and an astronomical almanac with the reported sighting.

The procedure followed in making final identifications deserves explanation because of the importance assumed by the identification as a basis for statistical treatment. As was mentioned, a conference of four qualified persons, two from ATIC and two from the panel of consultants, decided upon the final identification for each sighting report. This work was done at ATIC, periodically, as reports became ready.

During an identification conference, each sighting report was first studied, from the original data, by one person. If that person arrived at a decision, it was checked against the preliminary identification; if the two identifications were the same, the report was appropriately marked and considered finished. If the two identifications did not agree, the report was considered later by everyone participating in the conference until a group decision could be made.

If an evaluator was unable to categorize the report as one of the common

objects or as a natural phenomenon, and his opinion was that the sighting should be recorded as UNKNOWN, a group decision was also required on that report before it was considered finished. A group decision was necessary on all reports finally recorded as UNKNOWN, regardless of what the preliminary identification had been. In cases where a group decision was not made within a reasonable time, the report was put aside and later submitted to certain members of the panel of consultants for their opinions. If, after this, disagreement continued to exist, the report of the sighting was identified as UNKNOWN.

Upon completion of final identifications, all data were transferred to IBM cards, preparatory to analysis.

ANALYSIS OF THE DATA

Broadly stated, the problem at this point consisted of the judicious application of scientific methods of categorizing and analyzing the subjective data in reports of sightings of unidentified aerial objects. It was recognized that an approach to this problem could best be made by a systematic sorting and tabulation program to give frequency and percentage distributions of the important characteristics of sightings. A suggestion that an attempt be made to anticipate all questions that might be asked in the future about a sighting or a group of sightings, and to provide answers, was rejected. The systematic approach also made it possible to develop a detailed reference manual of the attributes of the sightings included in this study.

Thus, at the beginning of the analysis, a detailed plan was developed for sorting, counting, and tabulating the information from the punched-card abstracts of reports of sightings. It was believed at the time, and later substantiated, that the results of the program for sorting and tabulating would serve as a guide for the more sophisticated treatment involving statistical methods.

Also, it was anticipated that any patterns or trends that might be found could be subjected to concentrated study in the hope of discovering significant information relating to the characteristics of "flying saucers". Further, it was believed that these trends could serve as certain of the criteria of validity for any concepts (models) developed in the attempt to discover a class of "flying saucers".

The three parts of this study (1) the sorting and tabulation program, (2) the advanced study of the results of that program, and (3) the investigation of the possibility of conceiving a model of a “flying saucer” from descriptions reported, are discussed in sections entitled “Frequency and Percentage Distributions by Characteristics”, “Advanced Study of the Data”, and “The ‘Flying Saucer’ Model”.

FREQUENCY AND PERCENTAGE DISTRIBUTIONS BY CHARACTERISTICS

The original conception of this study assumed the availability of sufficient data to describe adequately the physical appearance, maneuver characteristics, range, direction, and probable path of the object or objects observed. However, familiarity with the data, acquired during the translation and transcription from reports to punched cards, indicated that there would be relatively few specific variables or factors that would yield meaningful correlation studies. Either the original data were too subjective, or the incompleteness of the original reports would seriously reduce the sample of a specific variable.

Preliminary tabulations of various sortings substantiated the impossibility of deriving statistical results from certain variables, such as movement of the observer during the sighting, sound, shape parameter, size, angular velocity and acceleration, appearance and disappearance, bearing, initial and final elevation, altitude, and orientation of the object. The statistically usable variables presented in this study include the date, time, location, duration, reliability, and method of observation of the sighting, and the physical attributes of number, color, speed, shape, light brightness, and identification of the objects sighted.

The presentation of frequency and percentage distributions of any of the variables must be interpreted in the light of the sample of incidents represented. For example, the analysis of the reported colors of the objects sighted, based on ALL SIGHTINGS, could lead to misrepresentation of the distribution of the reported color of the objects, because of the multiplicity of reports on some of the phenomena. On the other hand, the percentage distribution of the light brightness reported by each observer is more likely to be correct than a distribution based on one report for each phenomenon. To assure that the most nearly correct presentation was made, and to avoid the possibility of failure to uncover any pattern or trend

inherent in the data, the variables were studied on five different bases or samples. These samples, and their numerical relation to each other, were as follows:

- ALL SIGHTINGS (all reports) — 3,201 cards
- UNIT SIGHTINGS, all observers — 2,554 cards
- UNIT SIGHTINGS, single observer — 2,232 cards
- UNIT SIGHTINGS, multiple observers — 322 cards
- OBJECT SIGHTINGS — 2,199 cards

The preliminary tabulations indicated that the samples based on UNIT SIGHTINGS, single observer, and UNIT SIGHTINGS, multiple observers, would not add materially to this study. Accordingly, although the frequency distributions were recorded and are available for study, they are not presented in this report.

The bases of ALL SIGHTINGS, UNIT SIGHTINGS (referring to all observers), and OBJECT SIGHTINGS are presented in Appendix A as Tables A1 through A240. A critical study of these tabulations reveals that there is no apparent change in the distribution of any variable from one basis to another, and that no marked patterns or trends exist in any sample.

